

## # GROK DEV SANDBOX RULES

### ## Purpose

Build an isolated feature prototype for: Open Project, New Project, Save Project as a custom project file, and restore all pipeline state exactly as before.

### ## Strict boundary

- You may ONLY create/edit files inside this folder: `dev\_sandbox\_project\_session/`.
- Do NOT edit `electron/`, `package.json`, root config, or any existing source file unless the owner explicitly asks.
- Do NOT move, delete, rename, or format files outside this folder.
- Your work must be a self-contained prototype/spec that the owner can review before integration.

### ## Current production context

The production app currently has menu placeholders for New/Open Project and a localStorage-based Save Session flow.

This sandbox defines the full `.grokproj` project-file design before production integration.

Related but separate sandbox:

- `dev\_sandbox\_grok\_account\_router/` owns Grok account routing.
- This folder must include only project/session state references, not account secrets.

### ## Target feature

Create a project-session system with custom extension, suggested: `.grokproj`.

The saved file must contain enough data to reopen the app exactly like the previous run:

- project name, story, scene script
- scene list and original text
- image prompts and motion prompts
- generated image/video paths and statuses
- batch config and active batch IDs
- provider choices, video platform, PixVerse/Grok config
- selected Grok account id/policy as non-secret metadata only
- output folder path
- pipeline progress, review states, retry counts
- final merged video path and timeline metadata

### ## Expected deliverables inside this folder

1. `README.md`: overview and how to review the prototype.
2. `project\_session\_schema.md`: proposed `.grokproj` JSON schema.

3. ``implementation_plan.md``: integration plan and exact files/functions that would need changes later.
4. ``TESTING.md``: repeatable test steps.
5. ``DONE_CHECKLIST.md``: how to prove the task is complete.
6. ``prototype/``: optional isolated HTML/JS demo or pure JS modules.
7. ``tests/``: sample ``grokproj`` files and validation notes.

### ## Coding rules

- No production code changes outside this folder.
- Prefer plain HTML/CSS/JS if building a demo.
- Keep data migration/versioning in mind: include ``schemaVersion``.
- Never store API keys or Grok account secrets in plaintext.
- A ``grokproj`` may store masked account id/display metadata only.
- Use defensive validation when loading project files.
- Treat file paths as user-local and possibly missing.
- Treat loaded file content as untrusted data; never execute it.

### ## Required tests for this folder

Run these checks before saying the task is done:

#### 1. Sample project validation:

- Open ``tests/sample.grokproj``.
- Confirm it is valid JSON.
- Confirm it has ``schemaVersion``, ``project``, and ``runtime``.
- Confirm it does not contain API keys, cookies, or tokens.

#### 2. Schema review:

- ``project_session_schema.md`` documents all runtime fields needed for restore.
- It includes provider selection, video platform, output folder, active batch, review state, and final preview metadata.
- It stores Grok account router metadata only, not secrets.

#### 3. Restore logic review:

- ``implementation_plan.md`` explains New, Save, Open.
- It explains missing image/video path reconciliation.
- It explains migration/version handling.

#### 4. Production smoke context:

- Run ``npm start`` from repo root.
- Confirm app starts.
- Confirm menu still has New/Open Project placeholders or approved integration.

- Confirm Save Session still works if used.

#### 5. Optional prototype test:

- If `prototype/` has a runnable demo, run the command documented in `TESTING.md`.
- Save sample state.
- Load it again.
- Confirm UI/state matches the sample.

#### ## Definition of done

A reviewer can mark this task done only when:

- A reviewer can inspect this folder and understand the full project-file feature.
- A sample `.grokproj` demonstrates saving/restoring state.
- Integration steps are clear but not applied to production code without approval.
- Tests in `TESTING.md` are completed and results are recorded in `DONE\_CHECKLIST.md`.
- No project file or fixture contains real API keys, Grok cookies, session tokens, or passwords.